

**Efficiency of station connectivity and ease of transfer in Montgomery County,
Maryland's bus system**

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There has been a clear downward trend in ridership since the beginning of the 21st century (Shammout, 2021). Transit agencies have been facing declines in transit ridership despite growing populations, causing many issues with traffic congestion and emissions (Abbasi et al., 2021). There has been much study on potential ways to curb this decline and increase ridership, as well what the exact causes are.

There is no specific, defined field, process, or method of transit planning that is widely used (Khaled Shammout, 2021; Jarrett Walker, 2024; Owais, 2025). There is no specific process or software as an industry standard (Owais, 2025). This has positives and negatives; in countries like the United States of America, where transportation by private car is strongly emphasized and public transportation is underfunded, this has a negative consequence of different transit agencies taking very different approaches to the designs of their bus networks (Jarrett Walker, 2024).

Khaled Shammout suggests that a cause for ridership decrease is that transit agencies are doing what he calls “playing a finite game” and designing their systems “reactively”: transit routes are not built until it is known that people want to travel to that destination (2021). Jarrett Walker also describes the approach of designing transit systems around places where people “want” to go as being ineffective. Riders will go where transit goes; if there is a bus to an area, riders will go there, but if there is no bus, they will not travel there. He states that designing transit systems to increase access overall, without making this attempt to determine where people “want” to go, increases access to the region and can result in a more accessible and appealing transit system (2024).

Methods to improve ridership often involve bus network redesigns, sometimes including full system redesigns in transit systems seeing very low ridership (Rahman & Karner, 2023). There is also focus on specific areas of network design and transit features, such as scheduling, which is described as crucial for a system’s success and efficiency (Sevim et al., 2022; Altarazi & Janho, 2026). According to Altarazi & Janho, proper scheduling is absolutely vital for efficiency, and improper scheduling can increase wait time and trip time regardless of the presence of BRT features such as bus lanes. In their study, they use simulation to create an optimized BRT schedule (2026).

Jarrett Walker describes connective transit systems as the most effective network design. When transit systems are based around connections and changing vehicles to make a trip, it creates a system that is easier to understand, and reduces overall trip time (Jarrett Walker, 2024). At the same time, Ansari Esfeh, Saidi, Wirasinghe, & Kattan describe waiting time as being the greatest cost to riders, and names waiting time for transfers as a

part of this issue. In their study, they introduce a model to calculate the average total waiting time for all passengers (2022). My research takes a similar approach; I am proposing a model to calculate the success rate of catching a transfer to a “desirable”, connective station in 10 minutes or less after arriving at a given station.

Another contributor to ridership decline is the issue of the “first/last mile”. The first/last mile is the portion of a transit trip where riders are required to walk or otherwise transport themselves to a transit stop (Mohiuddin, 2021). The first mile, the travel from home to a transit stop, is important in context of transit in Montgomery County, and in context of my research. Because much of Montgomery County is suburban, many riders will have to take a bus from their neighborhood to a transit station before continuing their trip. If a rider lives in Germantown and works in Rockville, their regular trip is very likely to involve two or more stations, possibly starting with Germantown Transit Center. The connectivity that their first station provides to the rest of the system is crucial. As such, wait time and trip time issues resulting from scheduling, as well as the first-mile problem, are major priorities in my research and analysis.

In this analysis, I created a model that calculates the percentage of the time that a rider can arrive randomly at a given station and, within 10 minutes, catch a bus to the closest station with the most routes. The main goal is to create a method for measuring the efficiency of connectivity that one station provides riders in relation to the overall transit system. I believe that this can be achieved by calculating the percentage of random arrivals to a station that will have a short wait time for a bus to a nearby station that serves many routes. I do not intend for this model to be used completely on its own. It should be used

alongside other measures of bus access and connectivity. For this analysis, I expect to see a very low success rate of catching a bus overnight, and I expect that there will be many stations with a success rate of less than 0.8 during the weekday morning peak.

Research methodology and analysis

I used static General Transit Feed Specification (GTFS) from Montgomery County Ride On and WMATA Metrobus. The specific files I utilized throughout were the trips, stops, stop times, and routes. Stop times and trips were the most used files. In the stop time data, there are two time columns in HMS format: arrival time and departure time. These two

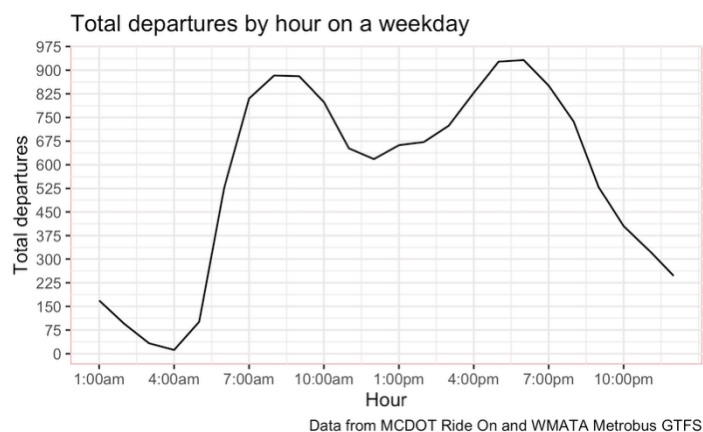


Figure 1: Total buses departing throughout the day on a weekday

columns are identical, but I will be using both words, “arrival” and “departure”, for ease of understanding. The datasets were joined and cleaned to show arrival times only at major bus stations and metro stations in Montgomery

County, Maryland: Bethesda Station, Forest Glen Station, Germantown Transit Center, Glenmont Station, Grosvenor Station, Lakeforest Transit Center, Medical Center Station, Montgomery Mall Transit Center, North Bethesda Station, Rockville Station, Shady Grove Station, Silver Spring Station, Traville Transit Center, Twinbrook Station, and Wheaton Station.

For my analysis, I created a model that checks every minute in a given time span for a bus departing a given station (“origin”) towards a predetermined station (“desirable destination”) within 10 minutes, and calculates the success rate by dividing the total minutes when there is such a bus available by the total minutes tested. As demonstrated in Figure 1, there are very few buses departing overnight; after about 10:00pm, there is a very sharp cut-off in bus trips, and at 4:00am, there are very nearly zero bus departures (buses stopping; not individual trips) in Montgomery County. There are also two peaks in this chart, in the morning, between around 6:00am and 10:00am, and in the afternoon, between about 3:00pm and 7:00pm. Considering the uneven distribution of trips, I will be splitting the time tested for rate of success in catching a bus into different time spans.

Before calculating the success rate, I find “desirable destination” for each station. This is done by calculating the distance between each station, filtering to the five closest stations, and selecting the station out of these which serves the largest number of unique routes. This method is imperfect for applications across diverse transit systems but works well in context of the set of stations and the transit system being analyzed. The threshold of five stations was chosen because this presented a variety of possible stations, and the distances between the stations was small. The station among the top five nearest stations which has the greatest number of unique routes was chosen because this station most likely provides the greatest reach; with the largest number of routes departing outwards from it, it is likely that riders will travel to this particular station to make a transfer. The transit system in Montgomery County emphasizes connectivity, and nearly all trips will include at least one transfer.

After selecting the desirable destination for each station, I created the model which calculates the success rate for catching a bus. The model takes each station and a time span. It increments one minute at a time until it reaches the end of the time span, and for each minute, the stop times data is searched for routes departing from the given (“origin”) station within one to ten minutes. The trip IDs associated with these routes are then searched in the stop times data to check if they have an arrival at the desired destination, and if the stop sequence for the desired destination arrival comes after the stop sequence for the origin station. For every minute that there is at least one such bus departing the origin station, the model counts a success, which is finally divided by the number of minutes tested to create the success rate. The specific range of one to ten minutes was chosen because this is considered within the range of acceptable waiting times for passengers waiting for transfer (Yang & Gao, 2025).

The time spans that I tested were morning peak (defined as 5:30am-9:30am), morning off-peak (defined as 9:31am-3:29pm), evening peak (defined as 3:30pm-7:30pm), evening off-peak (defined as 7:31pm-11:59pm), overnight hours (defined as 12:00am-5:29am), and day hours (7:00am-7:00pm). The model separately tested these time spans for the weekday schedule (Monday through Friday for Ride On, and Monday through Thursday for Metrobus), the Saturday schedule, and the Sunday schedule. The time spans would need to be adjusted for different transit systems; the specific hours chosen for peak and off-peak hours in this analysis were based around the schedule of the Ride On rapid transit bus route 101, which travels a major road in Montgomery County, Maryland, at peak hours only.

Findings

After calculating the probabilities, the stations with the highest success rate for the weekday morning peak were Lakeforest Transit Center, Silver Spring Station, Traville Transit Center, Wheaton Station, and Shady Grove station. The first four of these all had a success rate of 1.0.

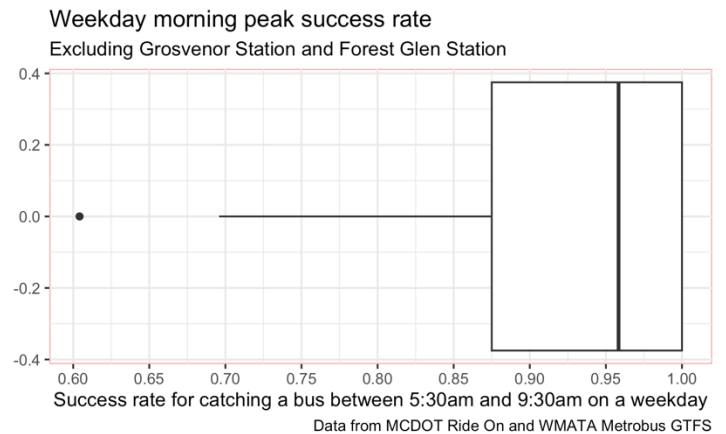


Figure 2: Success rate for weekday morning peak, minus Forest Glen

The five stations with the lowest success rates for weekday morning peak were Forest Glen Station, Twinbrook Station, Medical Center Station, Bethesda Station, and Montgomery Mall Transit Center. Forest Glen Station, which serves only two bus routes, had a success rate of 0.25, while Twinbrook, at the second lowest, had a success rate of 0.60. Montgomery Mall Transit Center had a success rate of 0.88. The average success rate was 0.85, and the median was 0.95. As seen in Figure 2, most of the success rates in the morning weekday peak were very high.

The mean absolute value of the difference between the morning peak and morning off-peak success rate was 0.08, and the median absolute value was 0.05. Bethesda station had a negative difference. North Bethesda Station, Medical Center Station, Glenmont Station, and Shady Grove Station had the largest changes in success rate from morning peak to off-peak, with each over 0.10. North Bethesda Station had a difference of 0.31. Figure 3 illustrates the success rate for morning peak and morning off-peak by station.

The mean success rate for weekday afternoon off-peak was 0.48, and the median was 0.54. Compared to morning peak, this was a mean difference of 0.37 and a median difference of 0.38. Going from weekday to Saturday afternoon off-peak, the mean difference in success rate has a mean of 0.08 and a median of 0.07.

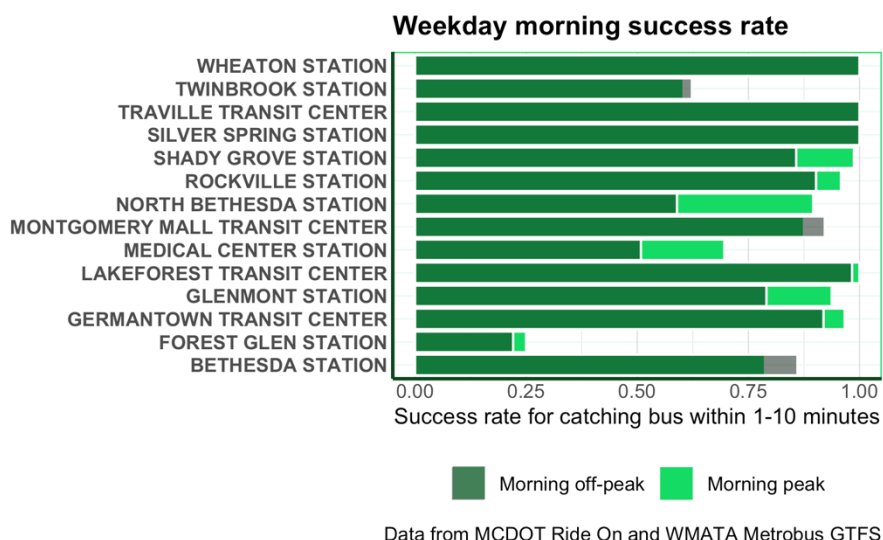


Figure 3: Weekday morning success rate, peak vs. off-peak

Going from weekday to

Sunday pm off-peak success rate, there was a mean difference of 0.08 and a median difference of 0.08. The mean success rate for weekday overnight hours (12:00am-5:29am) was 0.18 and the median was also 0.18. Grosvenor Station is being excluded from these results due to an issue with a special service route between the origin station and the desirable destination.

Overall, the results indicated better service than I had anticipated, and demonstrated a sometimes large difference in service, connectivity and convenience between different schedule times, especially later in the day. There are clearly areas for improvement within the system, such as improving non-peak service in stations such as North Bethesda Station and Medical Center Station. Improvement is important, especially considering the first-mile problem in transit. It is important that people are able to easily travel from their homes to their final destination, without losing too much time waiting.

Limitations

In this analysis, I only use transit stations within Montgomery County, Maryland. There were three transit stations in the Ride On stops data that I removed: Takoma Station, Takoma-Langley Transit Center, and Friendship Heights. These were removed because they are located outside of Montgomery County, but they are still close to different stations within the County, and are not irrelevant to the County's transit system. Further, I only used MCDOT Ride On and WMATA Metrobus, and not any passenger rail systems or other bus systems. As such, there is actually higher connectivity between many stations along the WMATA Red Line than the model's results imply. However, Metrorail was excluded due to the high fare, which makes it inaccessible for many riders, and makes it less likely that a rider will choose to use rail instead of bus to travel from one station to another nearby station. I also removed Grosvenor Station from my findings due to an issue with a special bus route.

Conclusion

Overall, the results from my model were not surprising, although they did indicate greater efficient connectivity than I had anticipated. It is clear that morning peak hours are very high efficiency and connectivity, and that catching a bus past midnight is extremely difficult, even at stations with otherwise very consistent success rates. I believe that my model, with some adjustments, could be a very useful model for measuring efficiency in connectivity across transit systems.

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